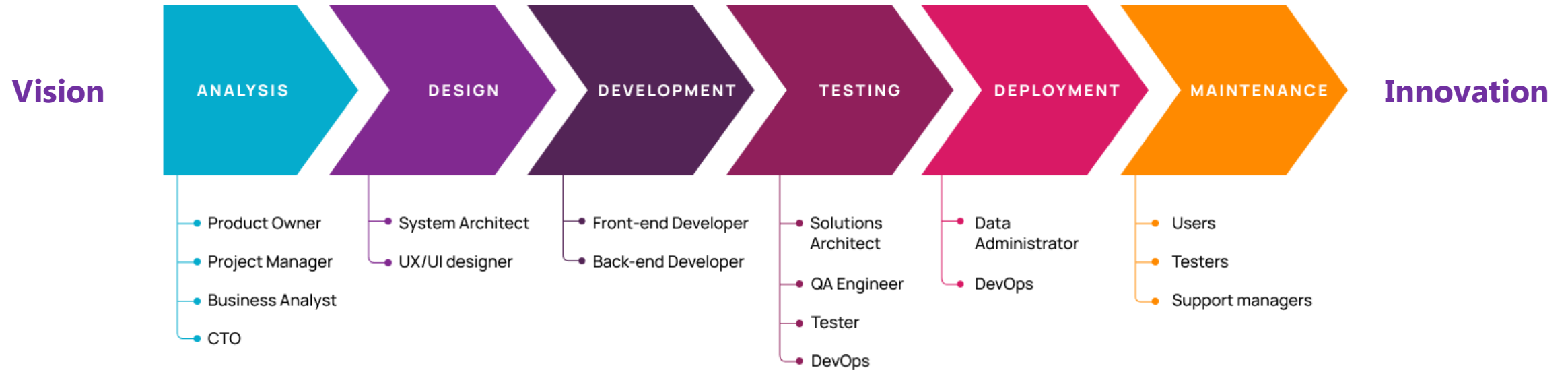


Technologies for E-Commerce

CAN302

Department of Communications and Networking
Xi'an Jiaotong-Liverpool University (XJTLU)

Lecture 12 – Create an e-commerce solution



What is a vision?



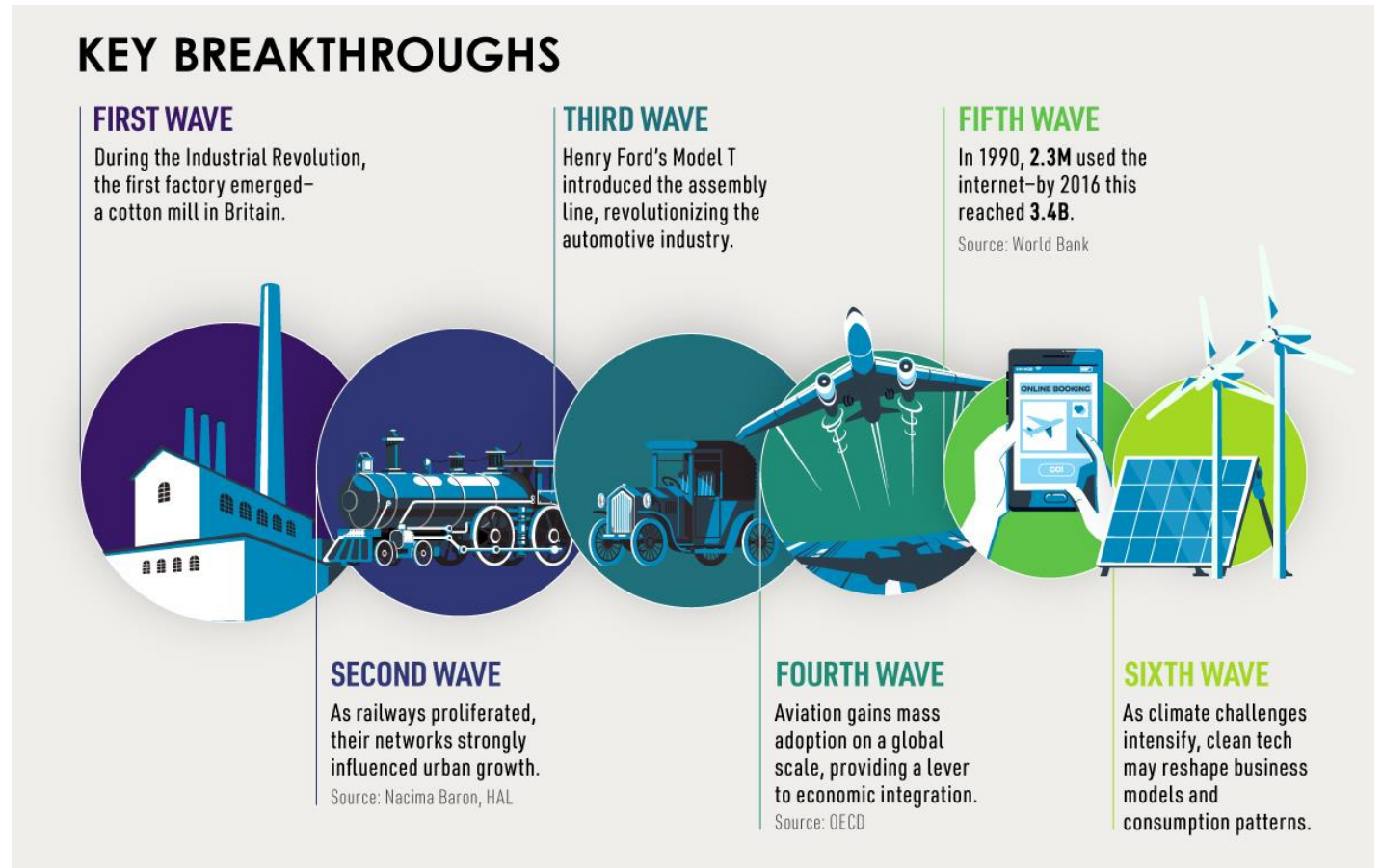
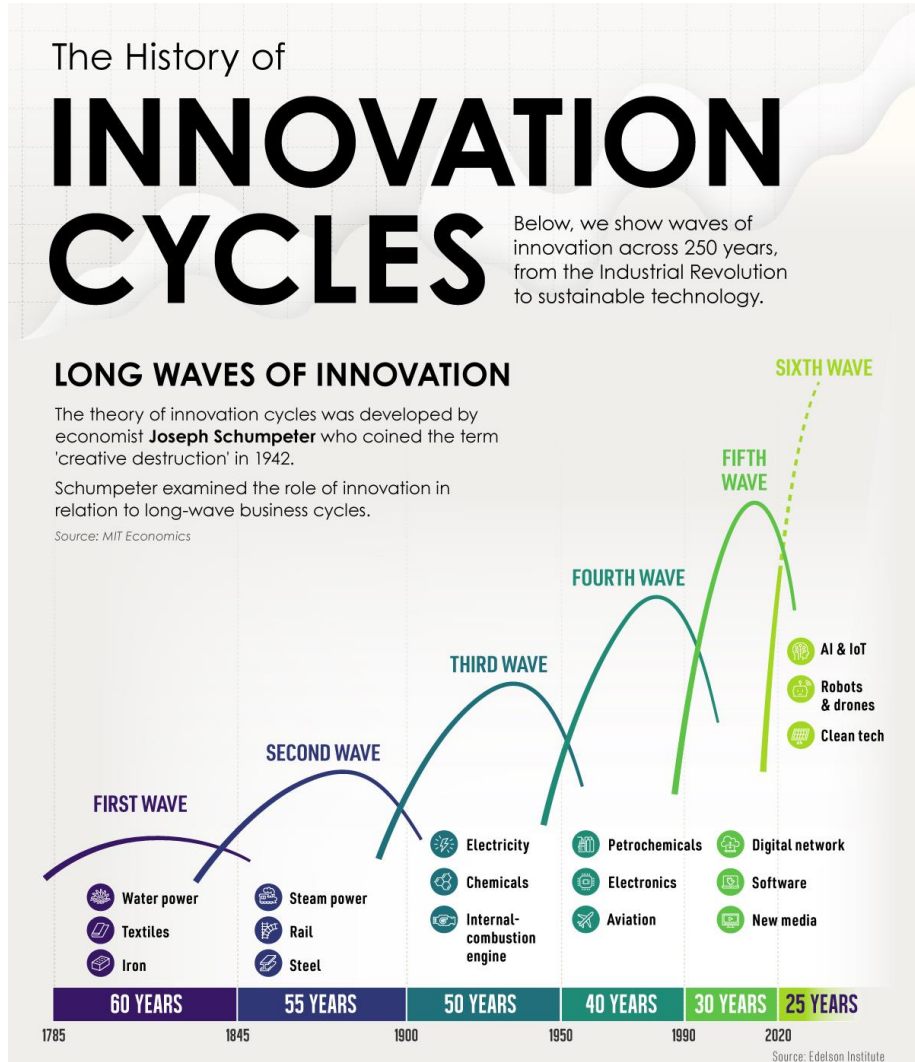
- Vision is the act of power of imagination that creates a mental picture of the future.

What is an innovation?



- Vision is the introduction of something new!

Innovation has cycles



Research and analysis



- Launching a business plan without research and analysis is like steering a ship without a **compass**.
- By researching industry trends and relevant markets, you're enabling your business to predict changes instead of simply responding to them.
- Professional research and analysis are important components in any **business plan**.

Data source

PRESS RELEASE
February 2, 2023

Apple reports first quarter results

Installed base crosses 2 billion active devices and hits all-time high for all major product categories

Services set new all-time revenue record



CUPERTINO, CALIFORNIA — Apple today announced financial results for its fiscal 2023 first quarter ended December 31, 2022. The Company posted quarterly revenue of \$117.2 billion, down 5 percent year over year, and quarterly earnings per diluted share of \$1.88.

- Reports by the public companies are important sources
- Every public company would publish the reports regularly.
- It is a reliable source to get many useful information.
- You can try to make a plan based on such numbers to judge the potential of your plan.
- You need to know them clearly before me **too!**

Research and Analysis



- Analysis always the step before design.
- Innovation always start from a vision.
- Vision could be a “**copycat**” type or **dream** one.
- The main different between copycat and innovation:
 - For “copycat” case, many success samples can be referred.
 - For dream case, it is more “following your heart”. **First** internet, **first** webpage, **first** web store, **first** smart phone ...

Copycat



Dream



Users' potential energy



- How different the new one different from the old one with similar function?

Users' potential energy

- Different environments would lead to very different potential and **barriers**



Activation Energy

Outcome of the research



https://wikimili.com/en/Business_plan

A business plan is a formal written document containing the **goals** of a business, the **methods** for attaining those goals, and the **time-frame** for the achievement of the goals. It also describes the **nature** of the business, **background** information on the organization, the organization's **financial** projections, and the **strategies** it intends to implement to achieve the stated targets.

In its entirety, this document serves as a **road-map** (a plan) that provides direction to the business.

Outcome of the research

Ecommerce Business Plan



- Executive Summary
- Business Overview
- Competitive Market Analysis
- Products and Offers
- Marketing Plan and Operations
- Financial Plan

<https://www.ecommerceceo.com/ecommerce-business-plan/>

Design

What is...

UX

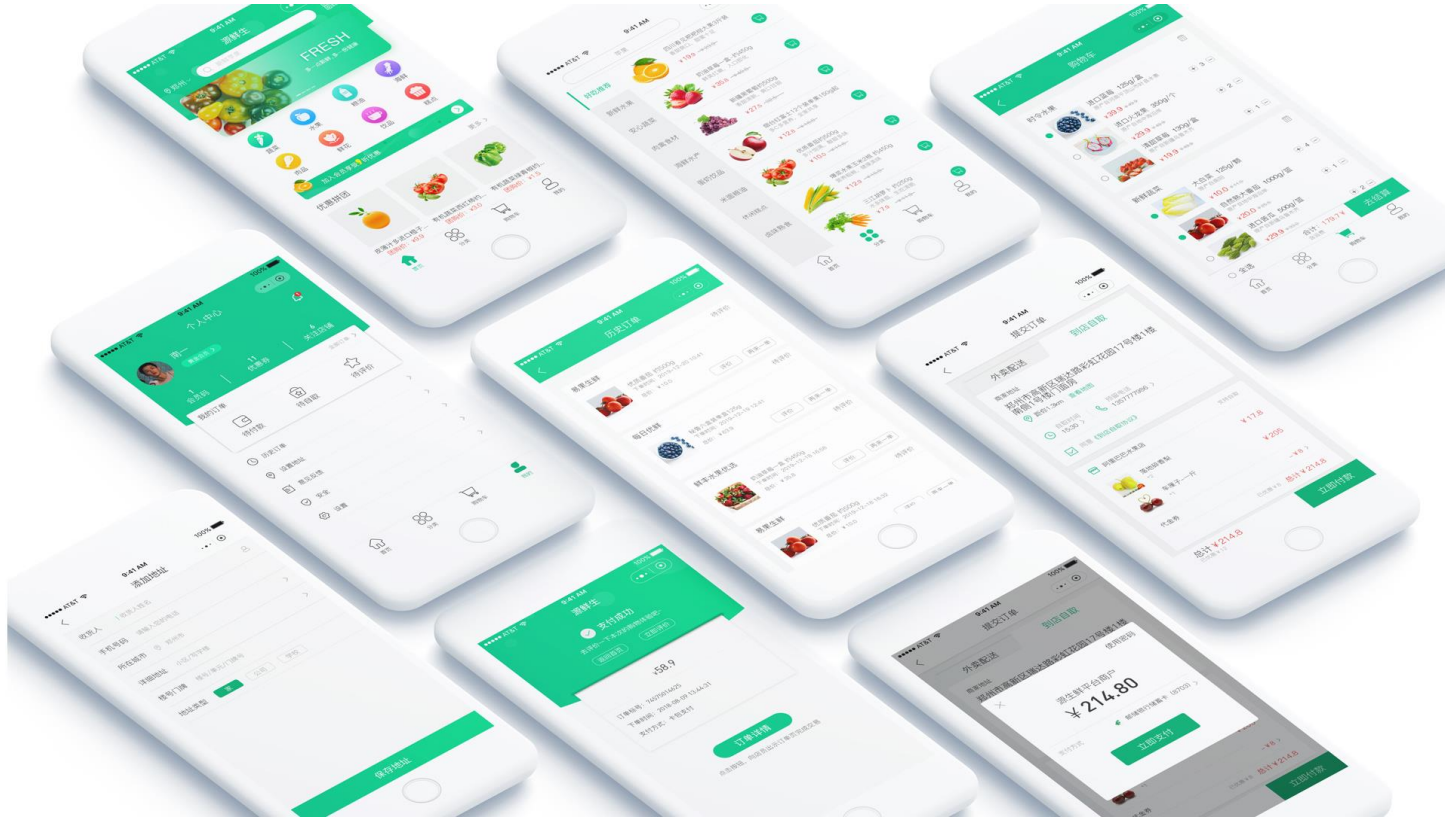
UI

IA

IxD

<https://www.zhihu.com/question/27928975>

UI: user interface



The user interface (UI) is the point of human-computer interaction and communication in a device. This can include display screens, keyboards, a mouse and the appearance of a desktop.

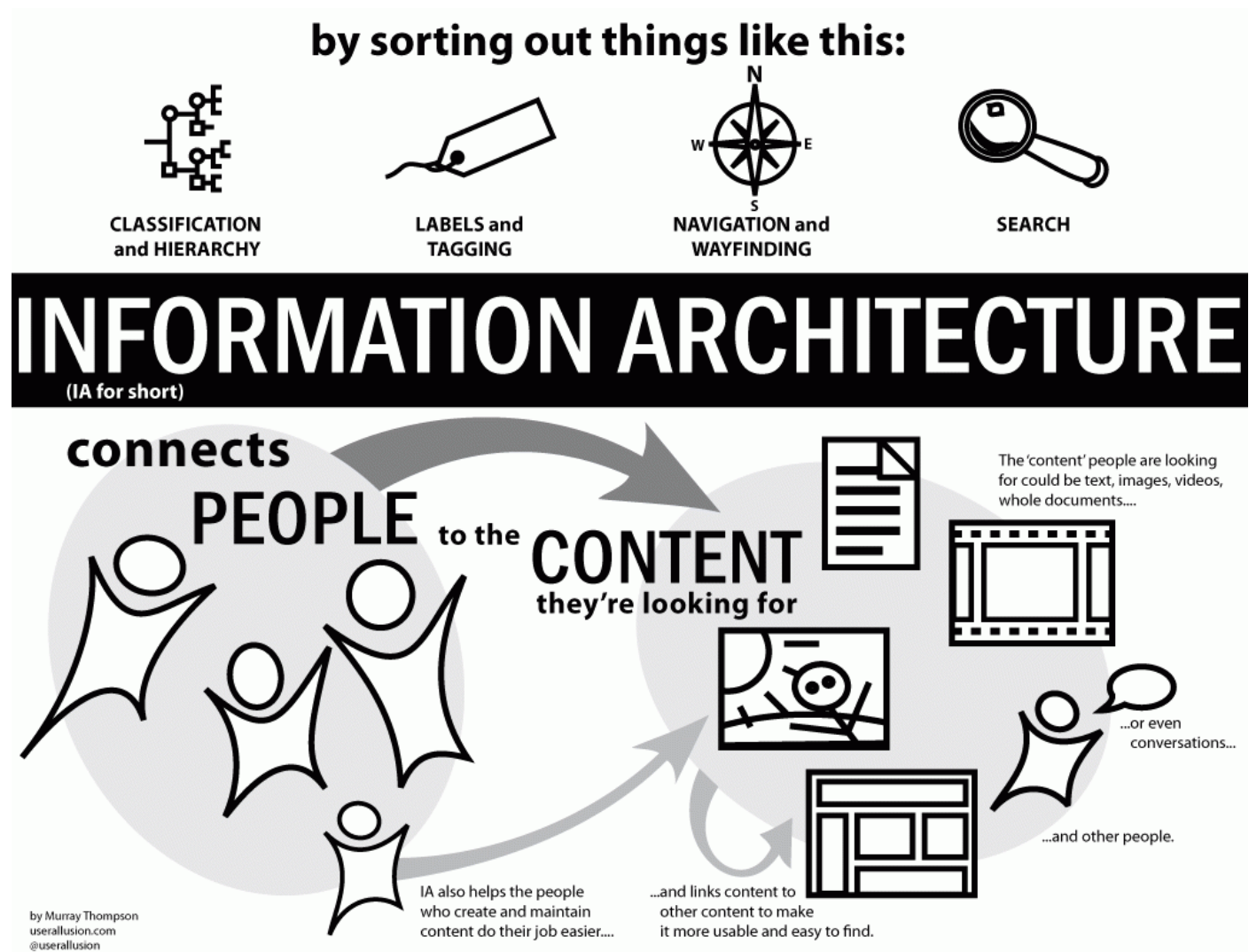
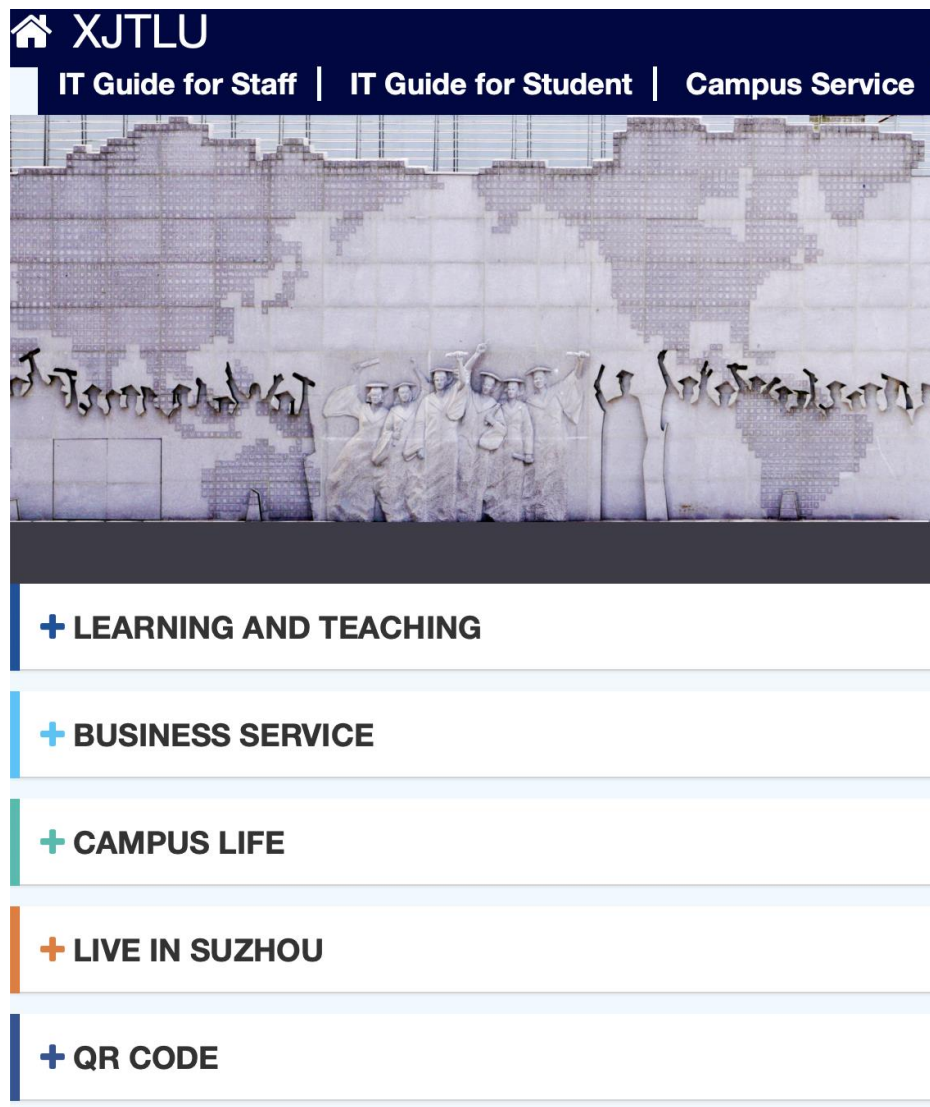
<https://searchapparchitecture.techtarget.com/definition/user-interface-UI>

IxD: interaction design



Interaction Design (IxD) defines the structure and behavior of interactive systems. Interaction Designers strive to create meaningful relationships between people and the products and services that they use, from computers to mobile devices to appliances and beyond.

IA: information architecture



<https://zhuanlan.zhihu.com/p/22168660>

UX: user experience

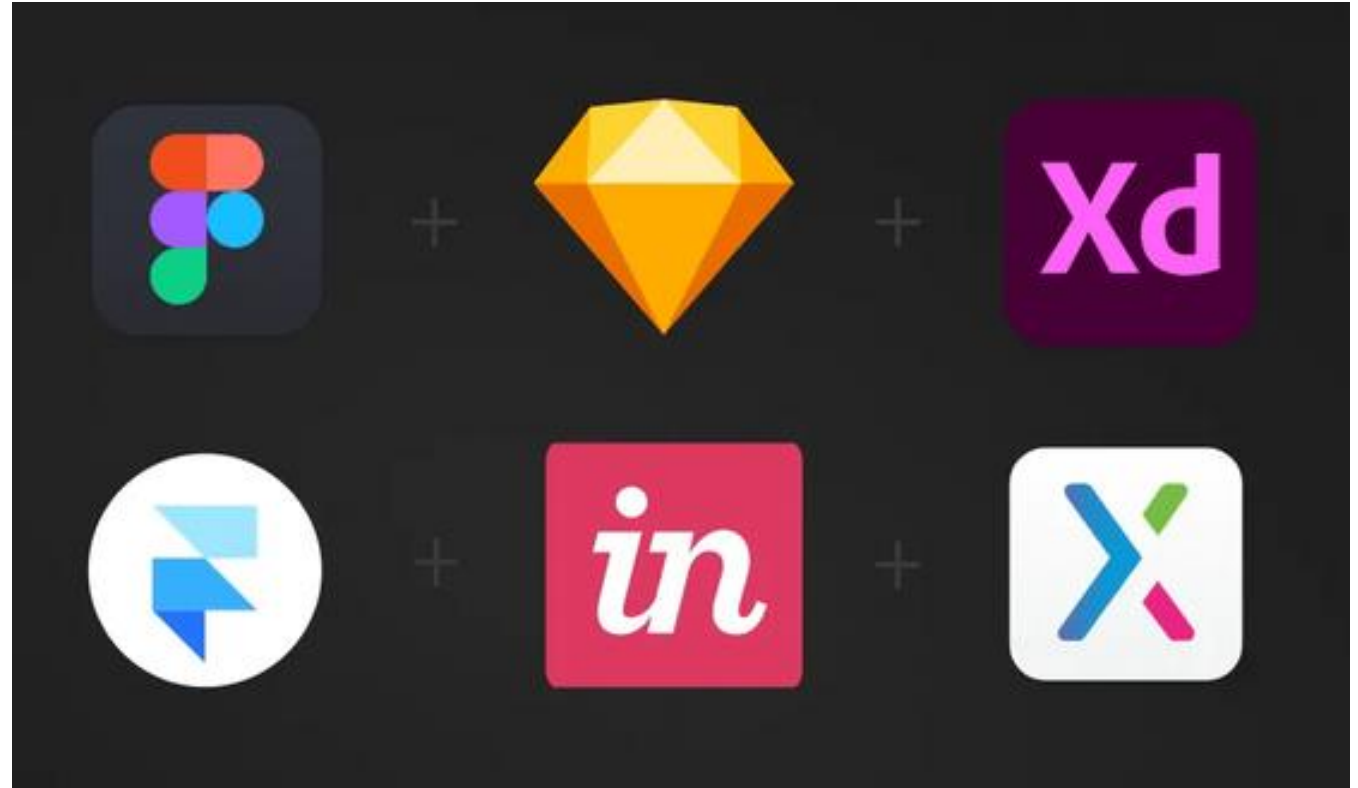


- In the digital design world, UX refers to everything that affects a user's interaction with a digital product. When people use a product, they usually [evaluate their experiences](#) according to the following criteria:
 - Value. Does this product give me value?
 - Function. Does this product work?
 - Usability. Is it easy to use?
 - General impression. Is it pleasant to use?
- UX is not solely, it is more like a lighthouse to guide the UI, IA and IxD

<https://xd.adobe.com/ideas/career-tips/what-is-ux-design/>

<https://www.interaction-design.org/literature/topics/ux-design>

Design tools

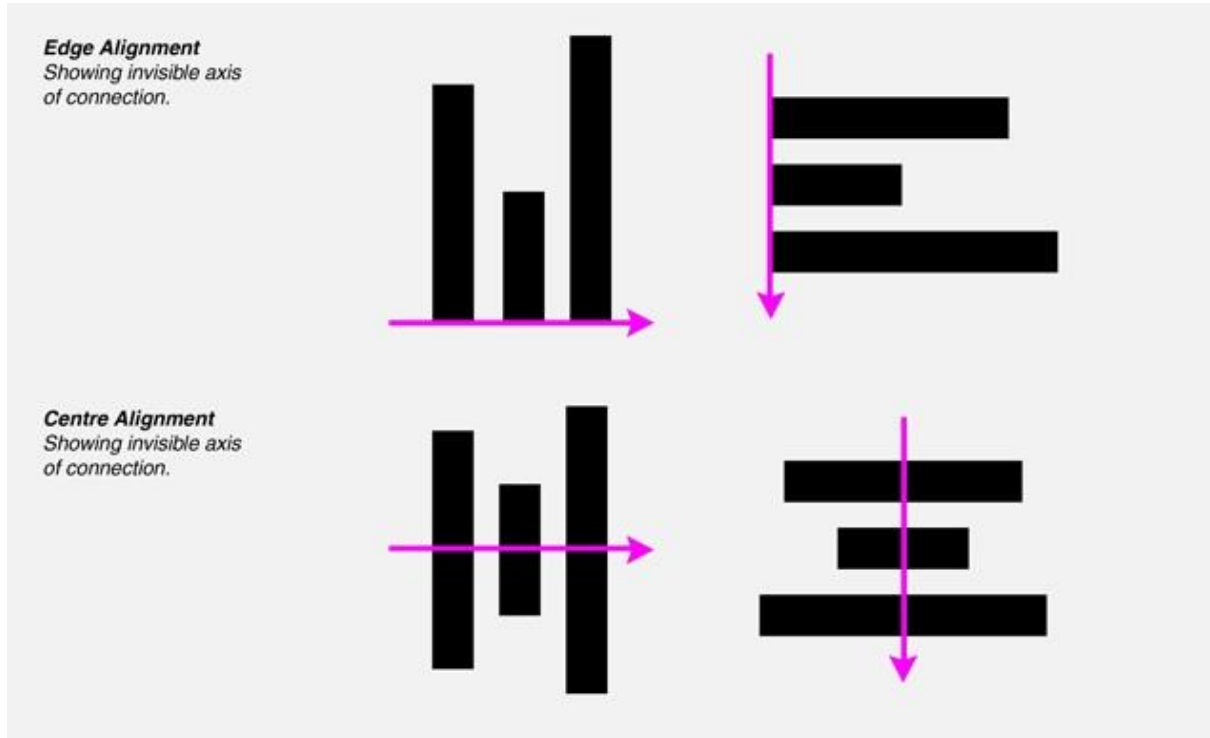


Sketch、Adobe XD、墨刀、Mockplus、Axure RP、Figma、Framer、Invision

<https://zhuanlan.zhihu.com/p/82214705>

<https://www.wendyzhou.se/blog/best-ux-ui-design-softwares/>

Tools are only tools



- Design isn't all about learning to use design software — although that's certainly important.

<https://www.freecodecamp.org/news/whats-the-difference-between-ux-and-ui-design-2ca8d107de14>

However, UI/UX is only a part of Design

Introduction ■ Purpose ■ Document Conventions ■ Intended Audience ■ Project Scope ■ References	Overall Description ■ Product Perspective ■ Product Features ■ User Roles and Rights ■ Operating Environment ■ Design and Implementation Constraints ■ User Documents ■ Assumptions and Dependencies	System Features ■ Description and Priority ■ Stimulus/Response Sequences ■ Functional Requirements
External Interfaces ■ UI/UX ■ Hardware Interfaces ■ Software Interfaces ■ Project Scope ■ Communication Interfaces	Concerns/Queries/Doubts ■ Performance ■ Safety ■ Security ■ Software Quality Attributes	Others ■ Limitations ■ Presumptions ■ Glossary ■ Analysis Models ■ Issue List

A product requirements document (PRD) defines the requirements of a particular product, including the product's purpose, features, functionality, and behavior. It serves as a guide for business and technical teams to help build, launch, or market the product.

PRD is not only the UI/UX, it should address 5 types of requirements:

- Business Requirements
- Market Requirements
- Functional Requirements
- Non-Functional Requirements
- UI/UX Requirements

You may not have best answers to all questions during the begging stage, but try to answer all of them.

<https://www.atlassian.com/agile/product-management/requirements>

<https://www.nimbleappgenie.com/blogs/how-to-write-the-product-requirement-document-for-your-mobile-app/>

<https://zhuanlan.zhihu.com/p/39744545>

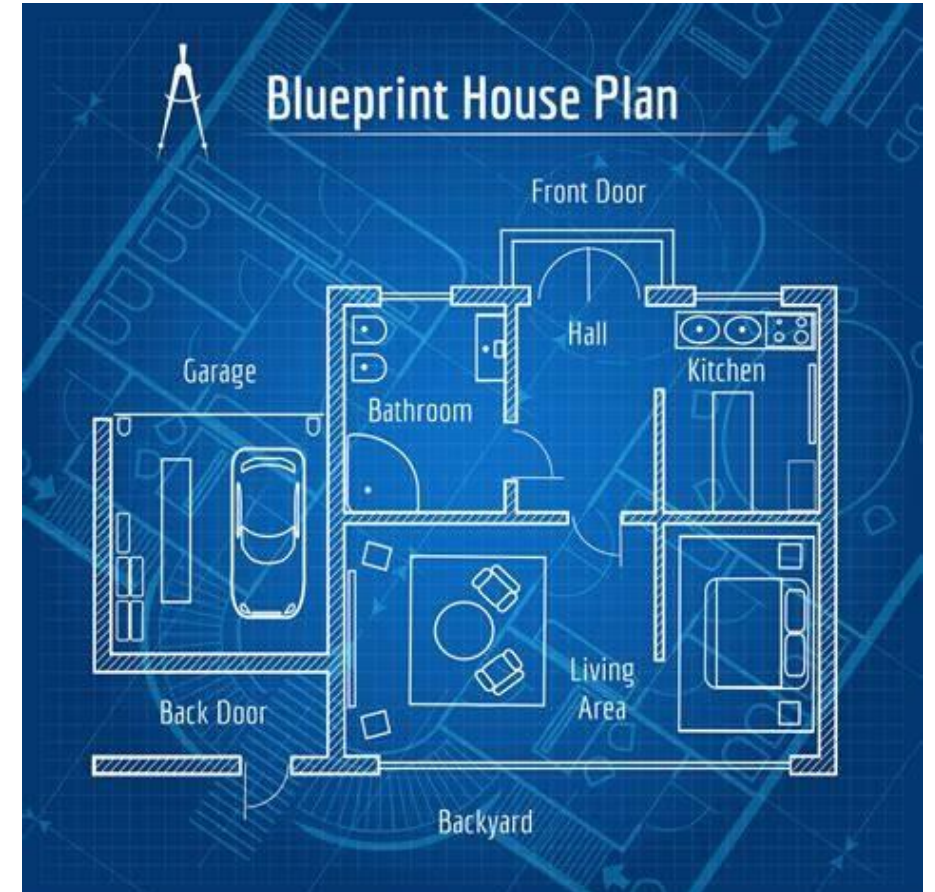
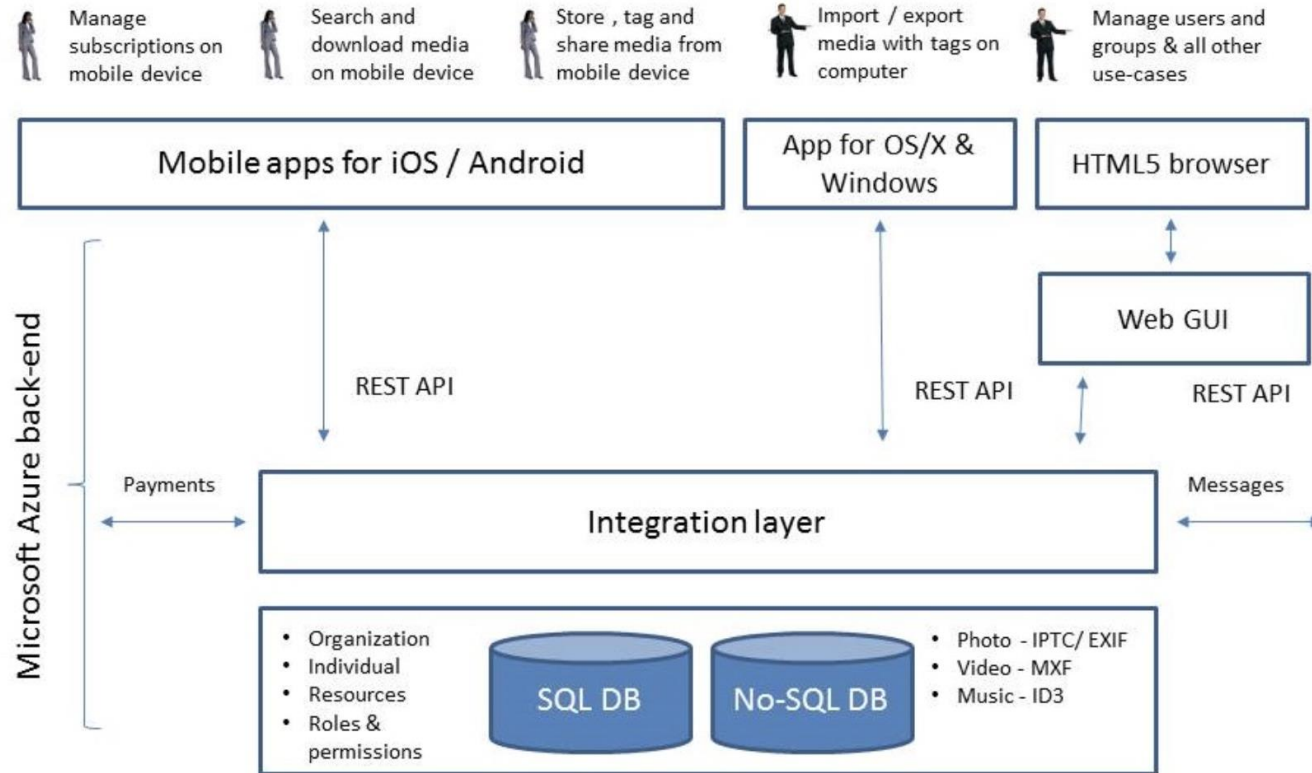
Example for non-functional requirements

Case of 12306

- Initial investment ¥300, 000, 000
- PV 150, 000, 000, 000
- Tickets per day 12, 000, 000

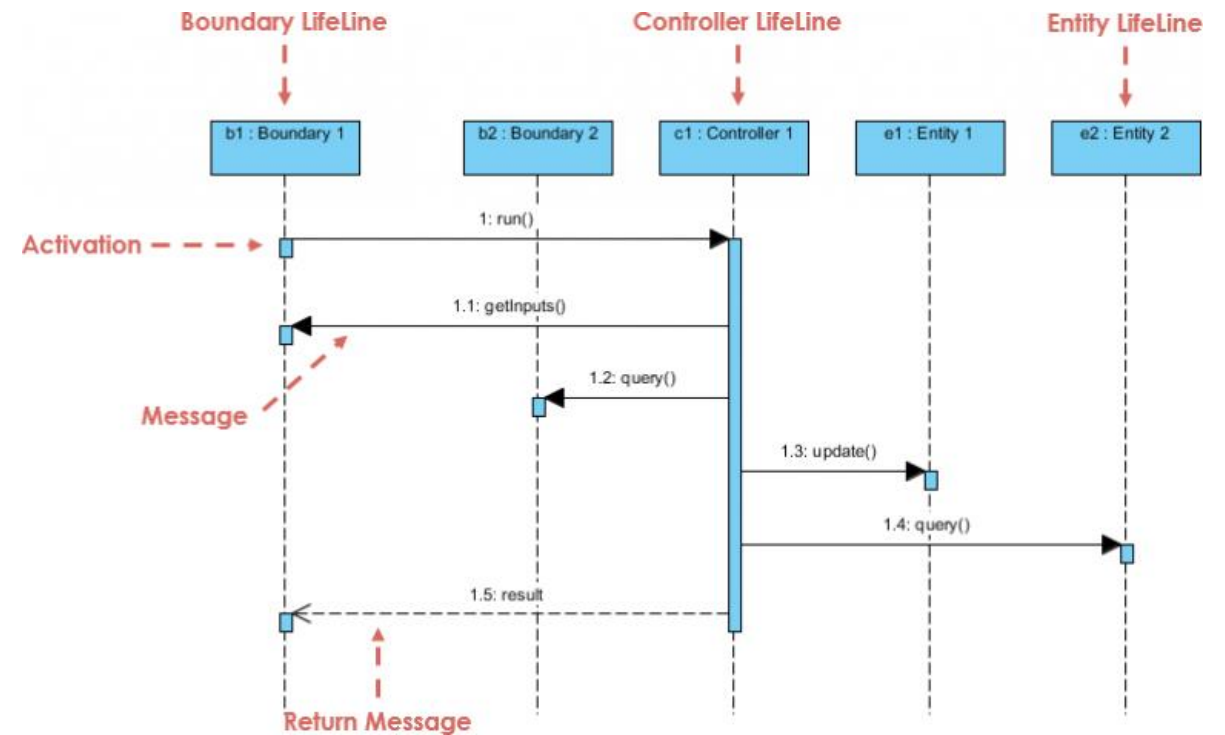
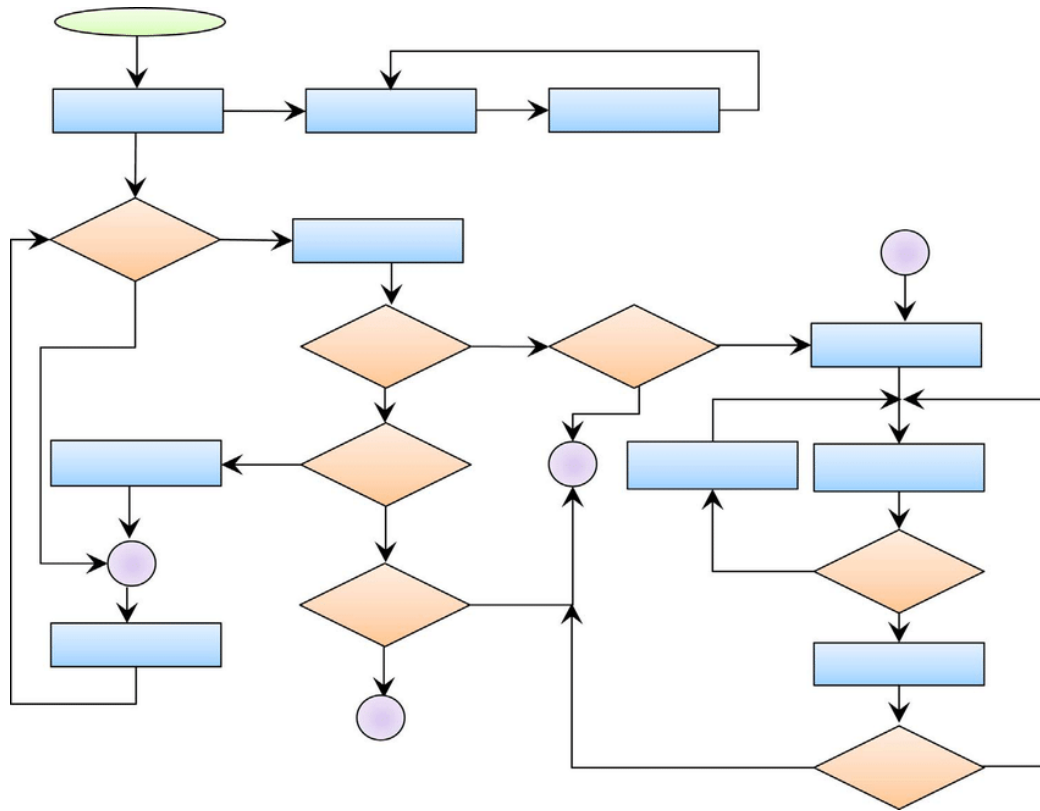


System architecture design



- A sample for **High Level design** (HLD).
- It works like the blueprint for a house.

Low level design (LLD)



- More details for each function.
- Many different diagram format: Flow Chart and UML

Develop

- While (earth is not destroyed) { }



- Write the code according to the design documents

996.ICU

English



996.ICU

What is "996"?

996 working, ICU waiting.

A "996" work schedule refers to an unofficial work schedule (9a.m. ~ 9p.m., 6 days a week) that has been gaining in popularity. Serving a company that encourages the "996" work schedule usually means working for at least 60 hours a week.



Outcome of develop



“Hello World!”

While many times...

工作三连

查bug



改bug



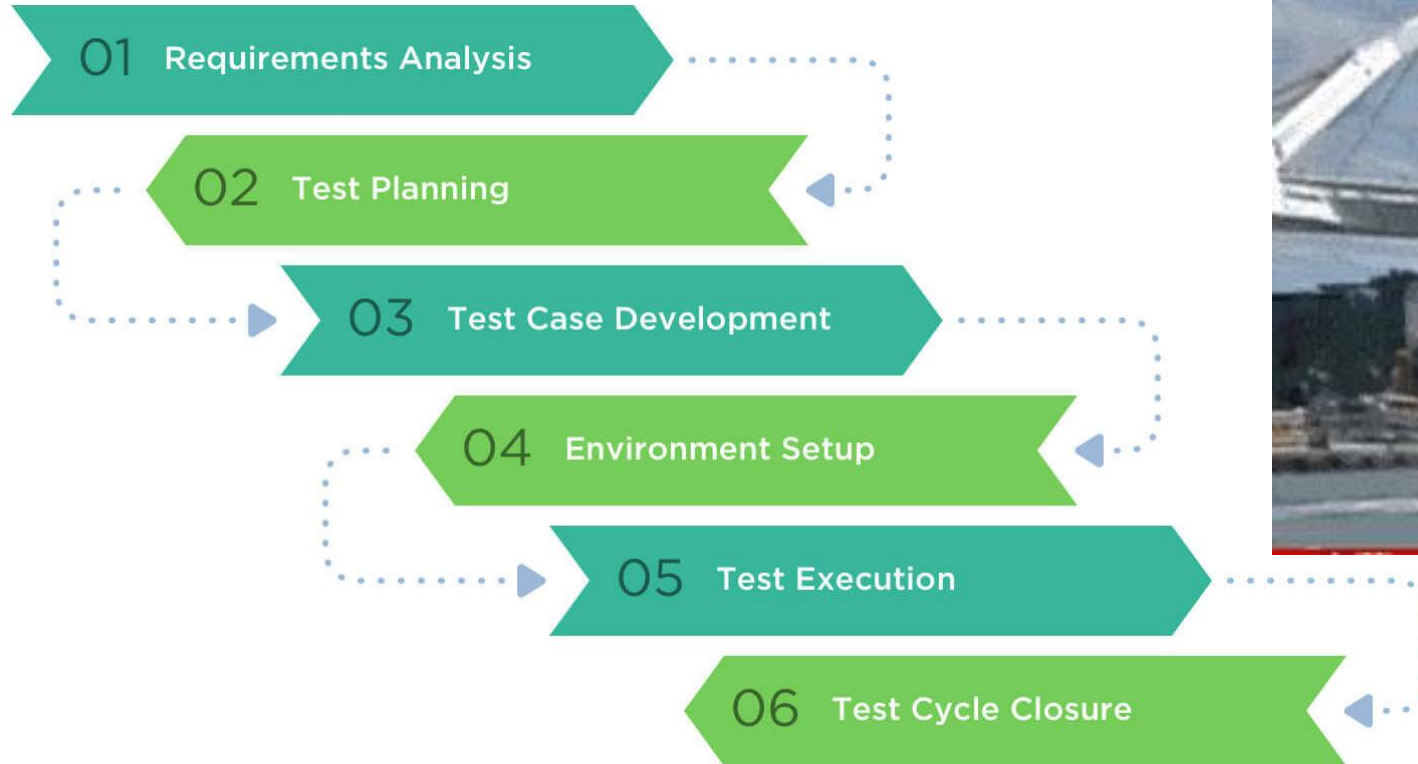
写bug



- Find bug, modify bug and write new bug...
- Finally, "bugs" make the whole thing work!



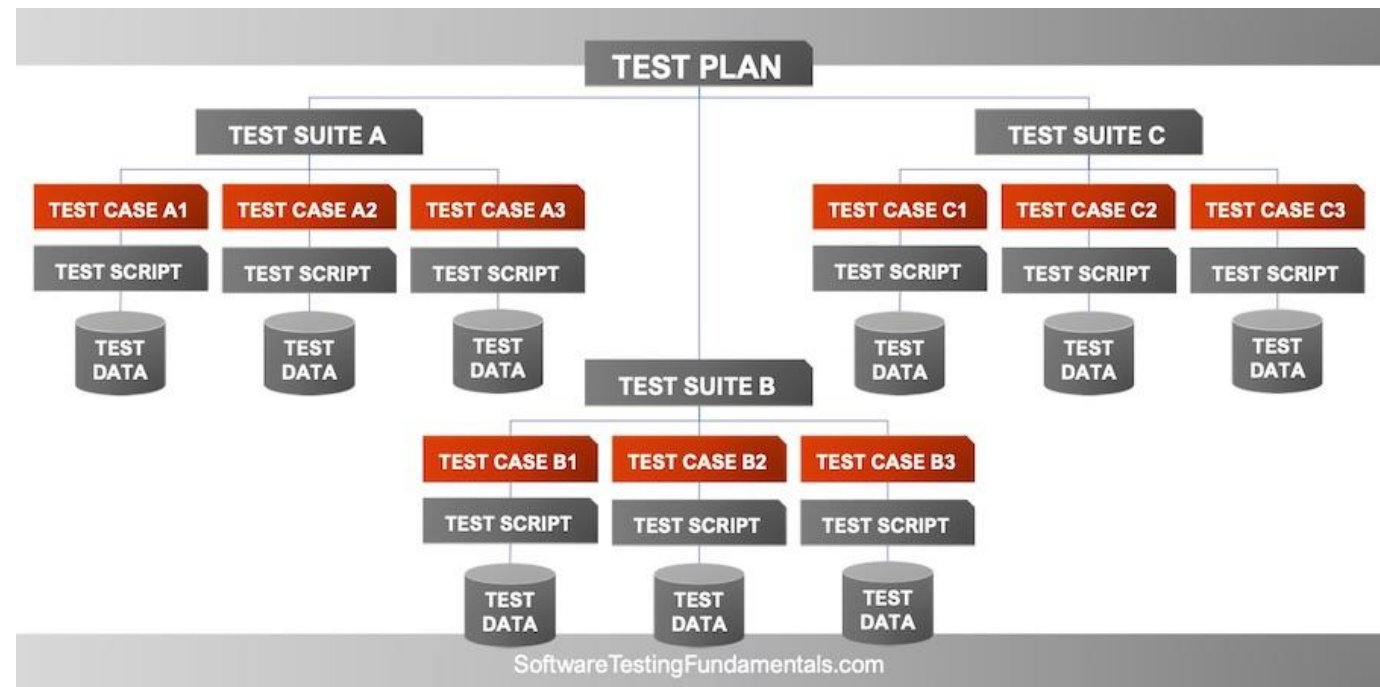
Test plays key role for the quality



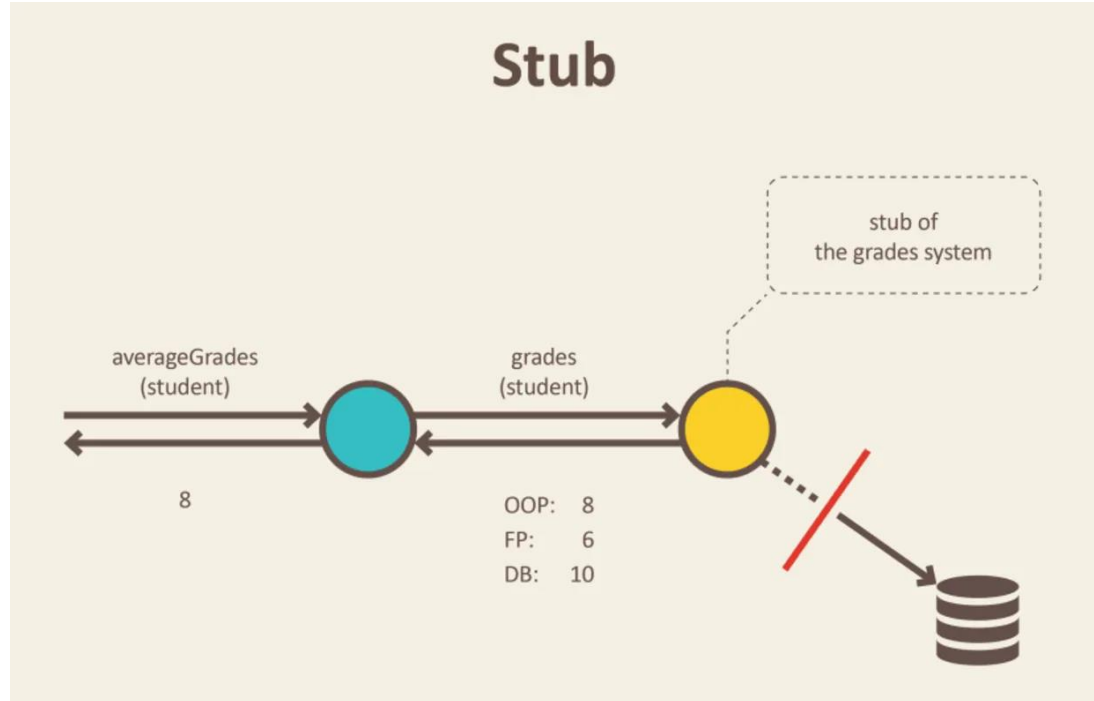
- Test should be included in the **design stage**
- Test should start at the beginning of coding

Test in different stage

- After the survey: write system test cases
- After design: write unit test cases
- After the module is submitted: execute the unit test case and revise it
- Unit test in progress: write integration test cases
- After module assembly: execute integration test cases and revise them
- Requirements change during development: modify all the above test cases
- Before release: execute system test cases



Fake, Stub and Mock



- Each unit needs input/outputs
- For the unit test, you need to “cheat it” with the simulated data as inputs
- Fake, stub and mock are similar terms for such simulation.

Test what

- Functional test: pay attention to whether the function is correct.
- Usability test: pay attention to whether the product is easy to use.
- Compatibility test: focus on whether the product is applicable to multiple platforms.
- Reliability test: pay attention to whether the product is stable and reliable.
- Security test: pay attention to whether the product has vulnerabilities.
- Performance test: focus on whether the product can run efficiently.

Test case sample

Summary	Steps	Expected Result	Functionality
Ensure the product list is properly laid out and rendered	1. Navigate to https://xxxxx 2. Click on the "Browse the catalog" link 3. Click on the "All products" link on the left navigation panel	The page is properly laid out according to the design	Product List
Ensure product can be searched by name	1. Navigate to https://xxxxx 2. Click on the "Browse the catalog" link 3. Enter some keyword into the Search textbox in the header and click Search	Products listed are partially matched with the keyword entered	Product List
Ensure no product returned when searching with special characters	1. Navigate to https://xxxxx 2. Click on the "Browse the catalog" link 3. Enter "!@#%&*()" into the Search textbox in the header and click Search	No product is listed in the product list	Product List

- Test result only could be Pass or Fail

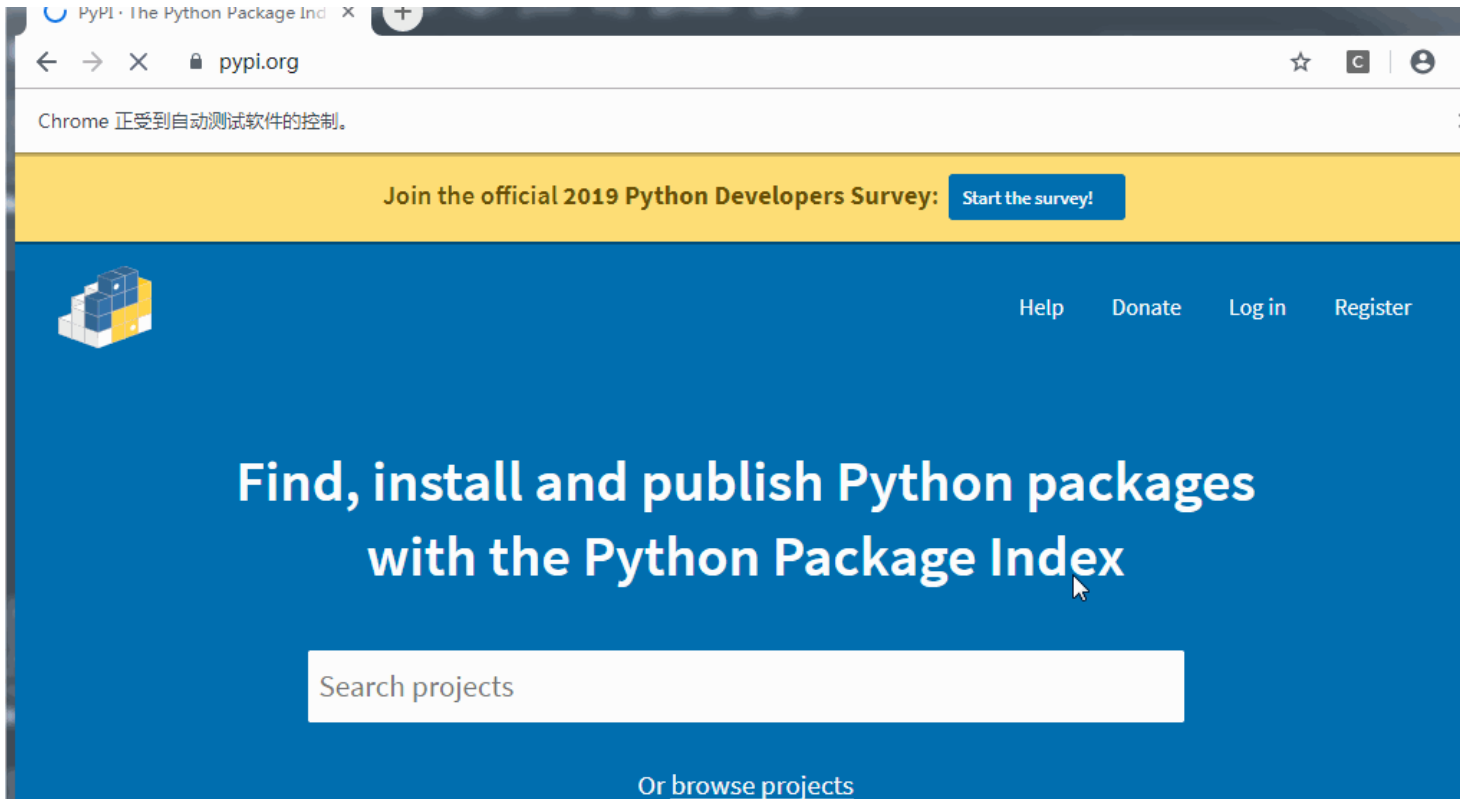
Browser compatibility means ...

	CHROME	SAFARI	IE & EDGE	FIREFOX	OPERA
	 57 %	 13 %	 11 %	 9 %	 3 %
  ANDROID 40 %	✓	--	--	✓	✓
  WINDOWS 36 %	✓	✗	✗	✓	✓
  iOS 13 %	✗	✗	--	✗	✗
  OS X 5 %	✓	✓	--	✓	✓
  LINUX 0,7 %	✓	--	--	✓	✓

- Manually test would be a nightmare.

- **One might wonder – why does my website look different on different browsers? Well, the answer is simple.**
- Each browser has distinct rendering engines. (Chromium-based browsers like Google Chrome & Microsoft Edge use Blink whereas Safari browser uses WebKit)
- Each engine processes the code written for other websites in its own unique way.
- Browsers like Chrome, Safari, and Edge often interpret or display website source code (HTML and CSS) differently.
- Since each browser's engine will interpret and render a webpage distinctively, the same website can look, feel, and function differently across browsers.
- The visitor's OS also plays a key role in how the website looks different on different browsers.

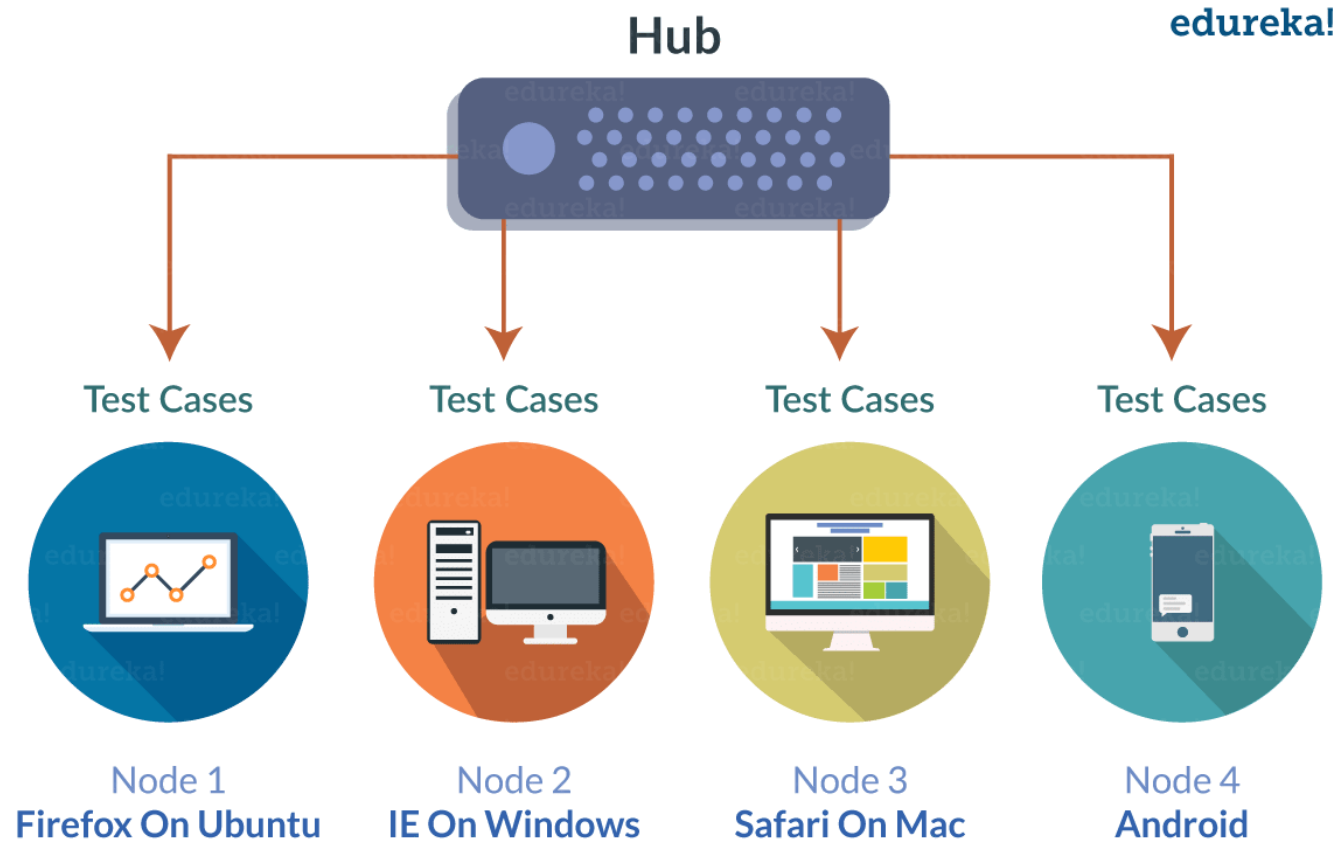
Web test automation by *selenium*



```
1 from selenium import webdriver
2 from selenium.webdriver.common.keys import Keys
3
4 driver = webdriver.Chrome()
5 driver.get("https://pypi.org/")
6
7 print(driver.title)
8 assert 'Python' in driver.title
9
10 elem = driver.find_element_by_name('q')
11
12 elem.clear()
13 elem.send_keys('deniro')
14 elem.send_keys(Keys.RETURN)
15
16 assert 'No results found.' not in driver.page_source
17
18 driver.close()
```

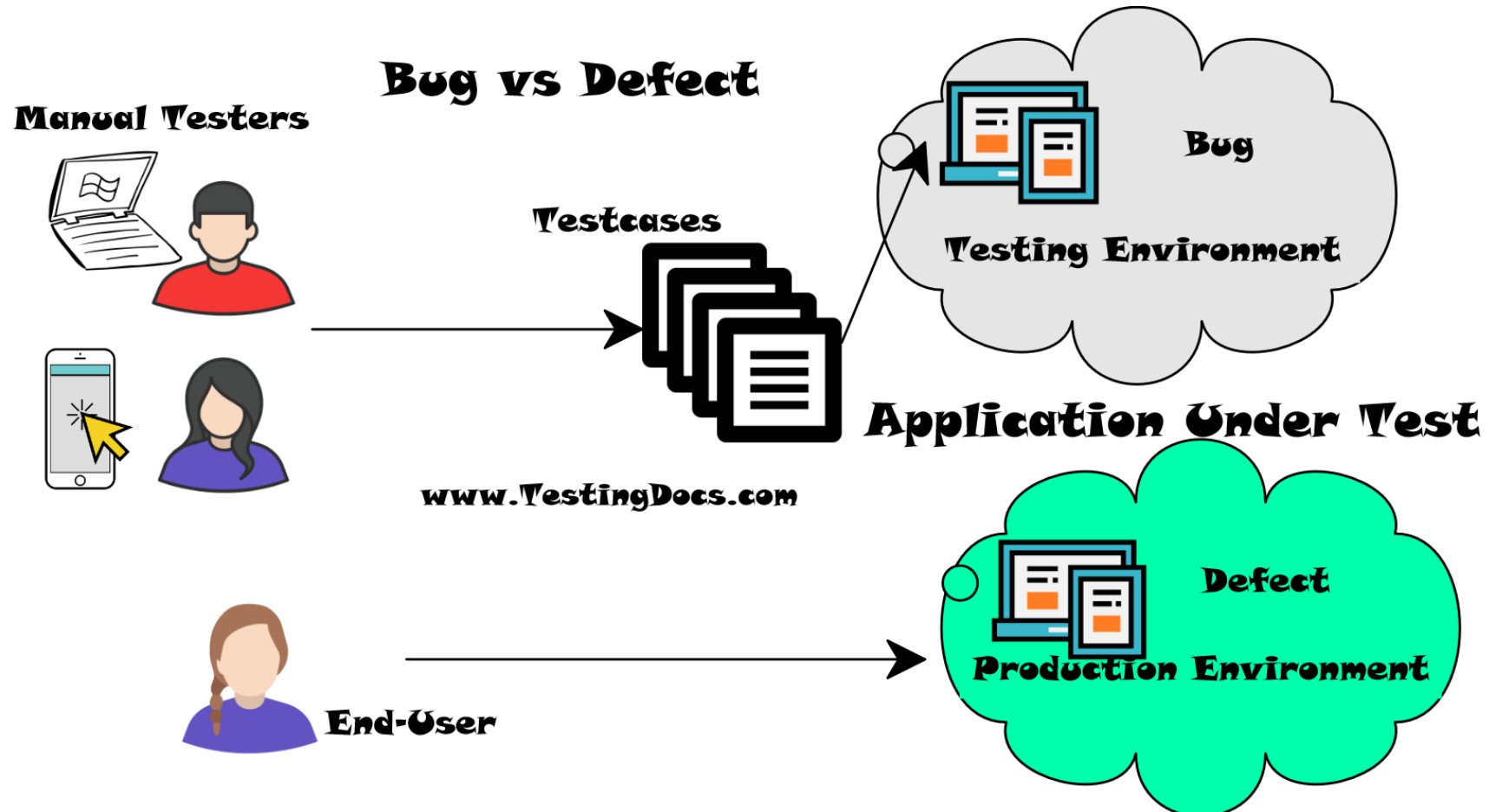
- Selenium is a driver that you can operate DOMs by code!

Selenium Grid



- Using Selenium Grid, different OS/Browser combinations can be automatically tested.

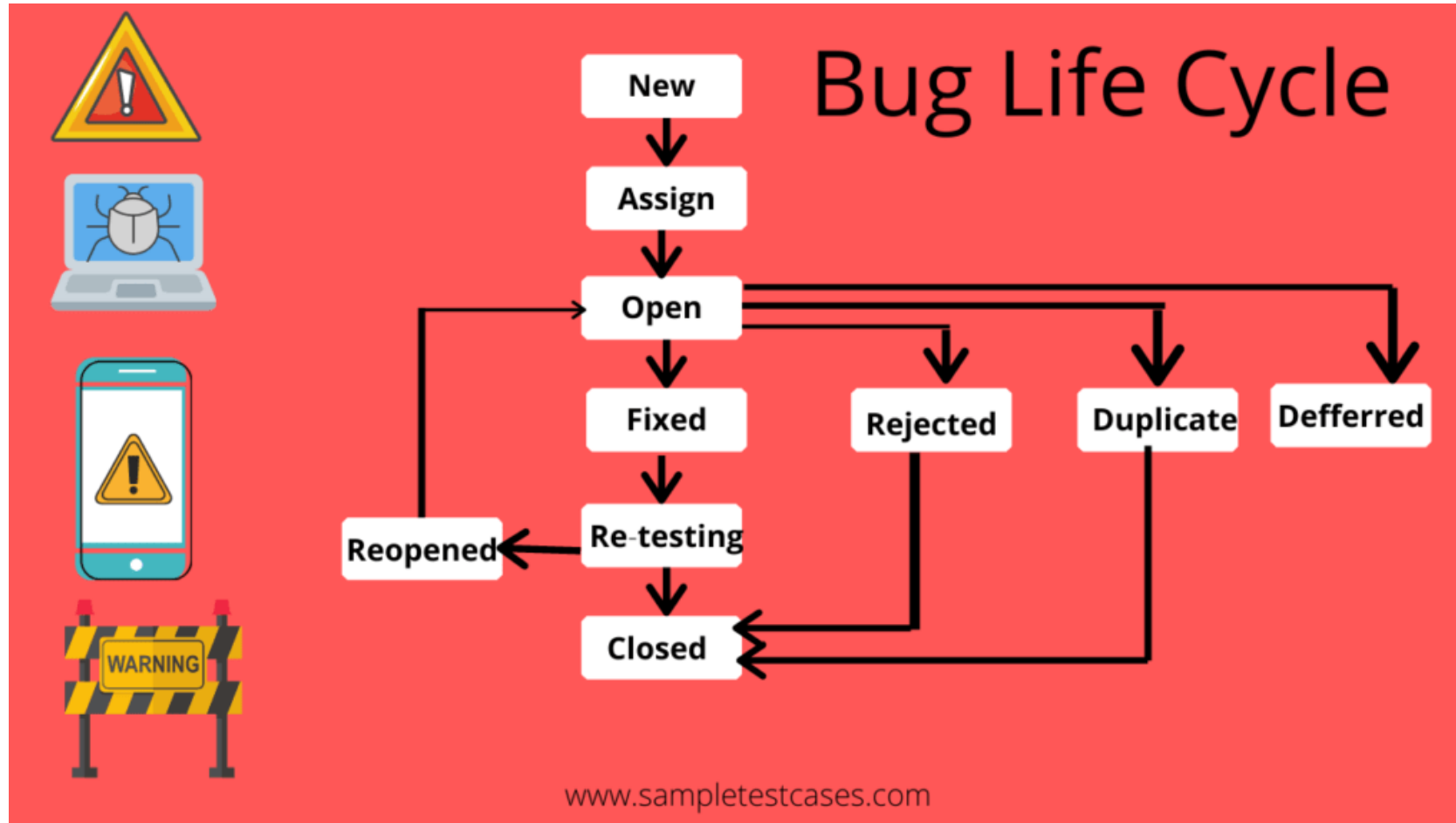
Bug and defect



- Not possible to cover all possibilities

<https://www.testingdocs.com/questions/what-is-software-bug-defect/>

Bug life cycle

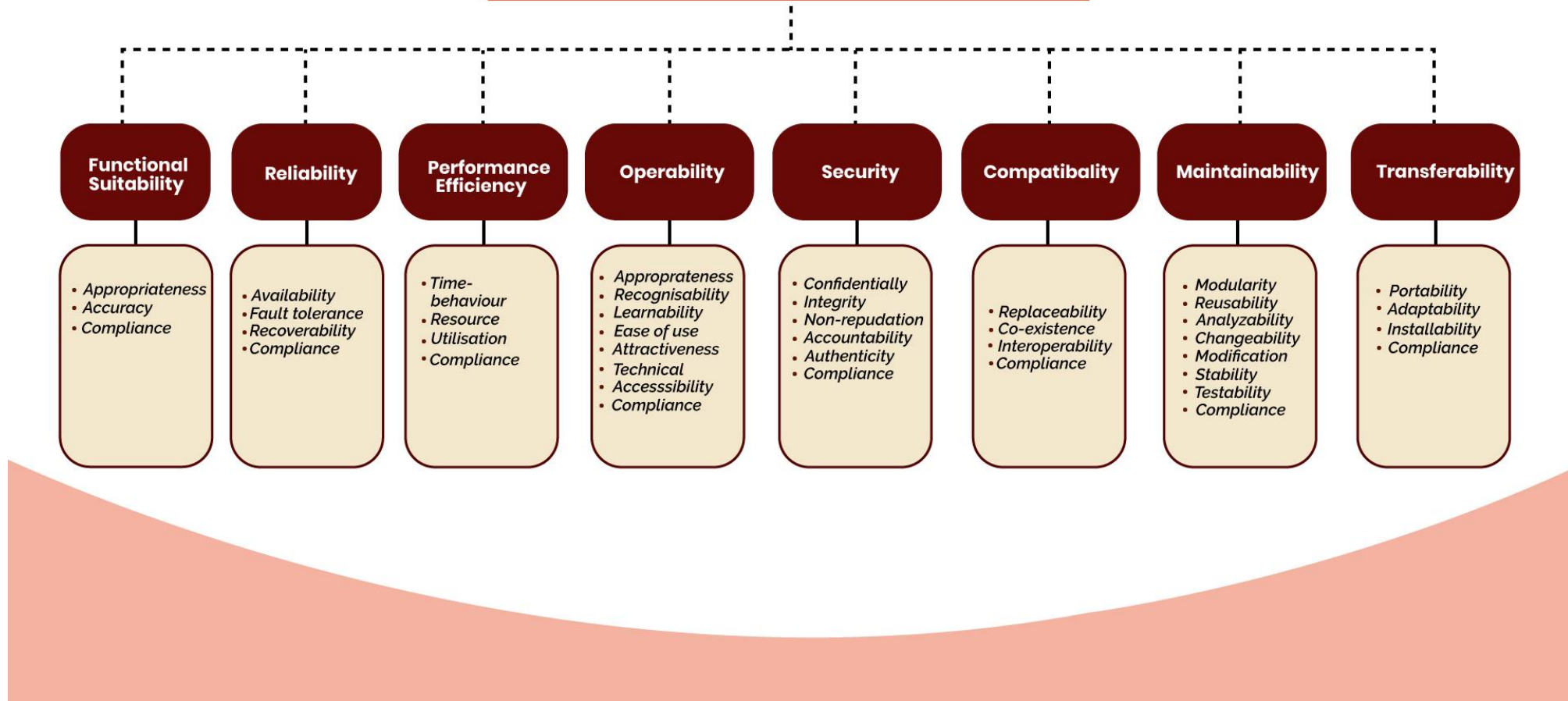


- Not all bug will be fixed
- But at least, you know it will happen

<https://sampletestcases.com/bug-report/>

ISO/IEC 25010:2011 Software Quality Model

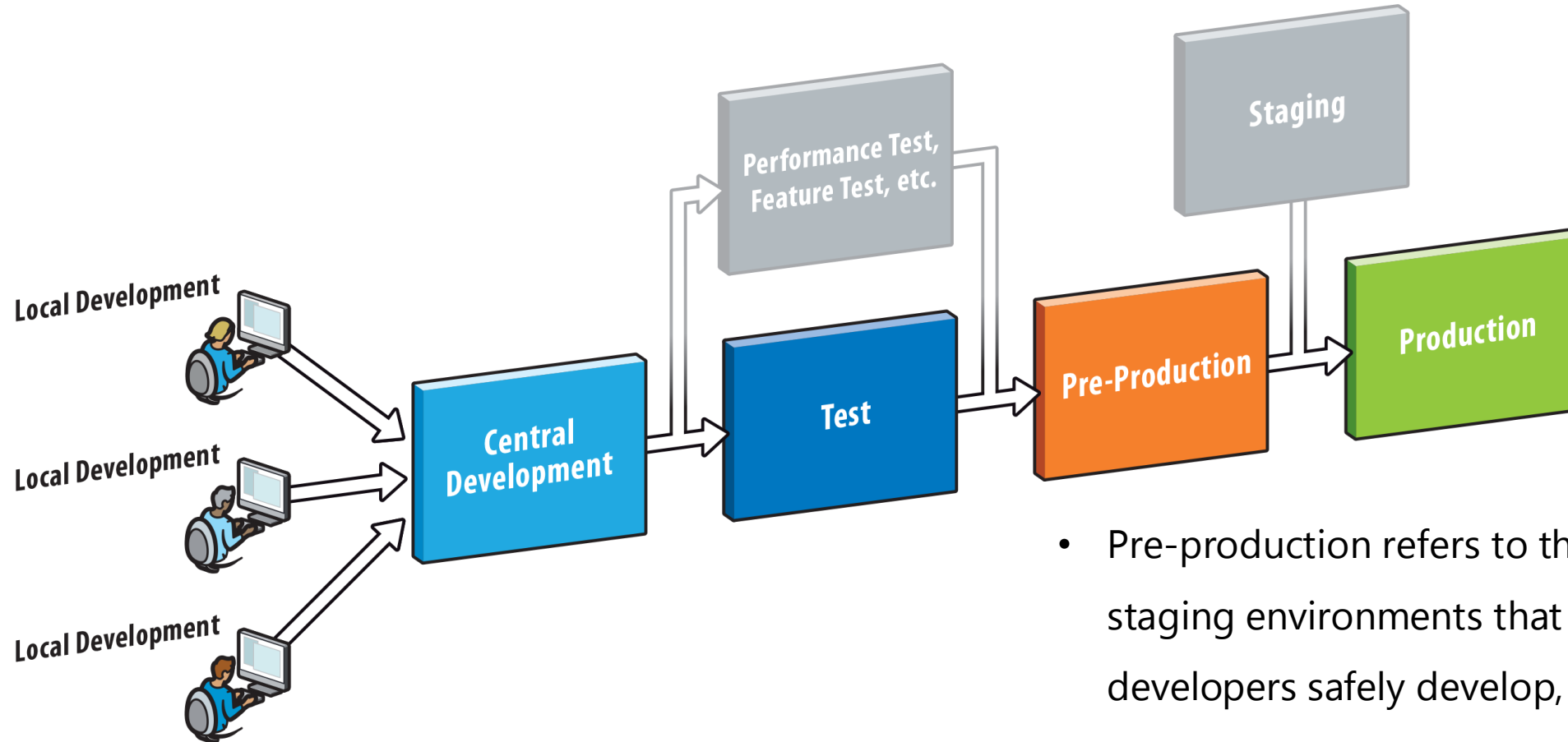
Software Product Quality



Implement or Operation



Different environments

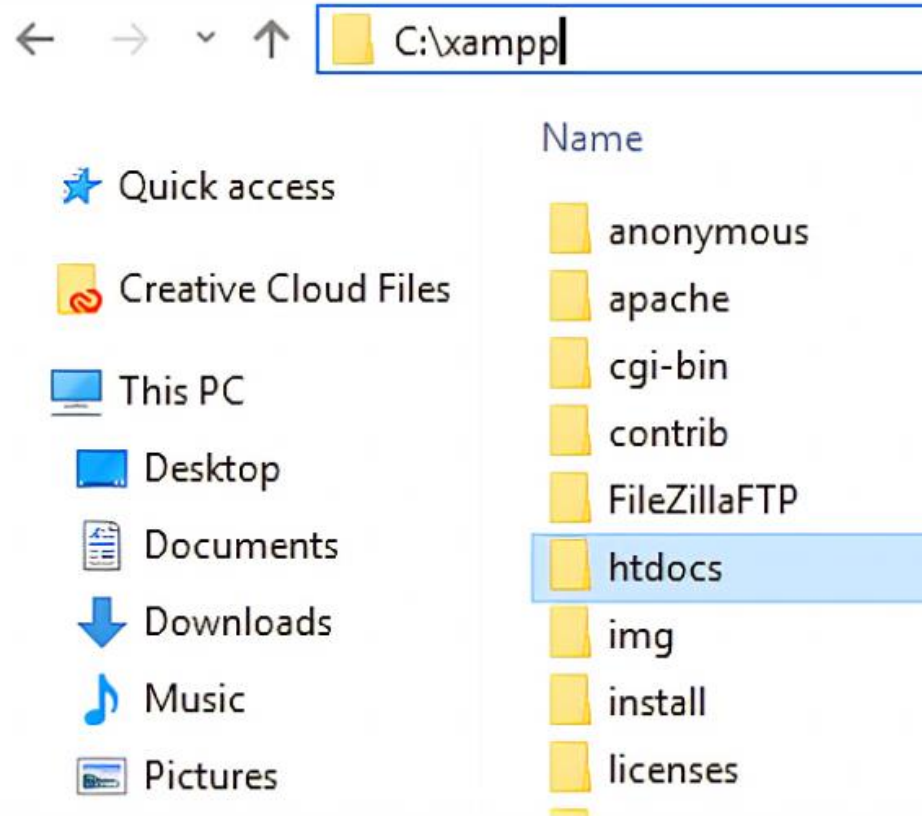


- Pre-production refers to the various staging environments that help developers safely develop, test and integrate new code.

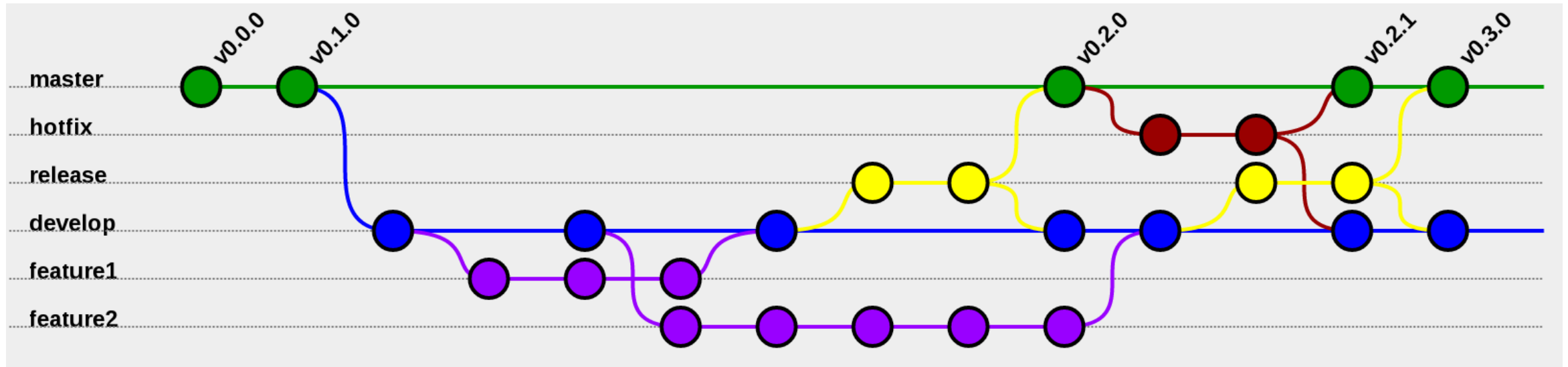
<https://www.jianshu.com/p/ba20b064fd25>

<https://thenewstack.io/pre-production-environments-key-concepts-and-best-practices/>

Implementation: “Copy” code to web server

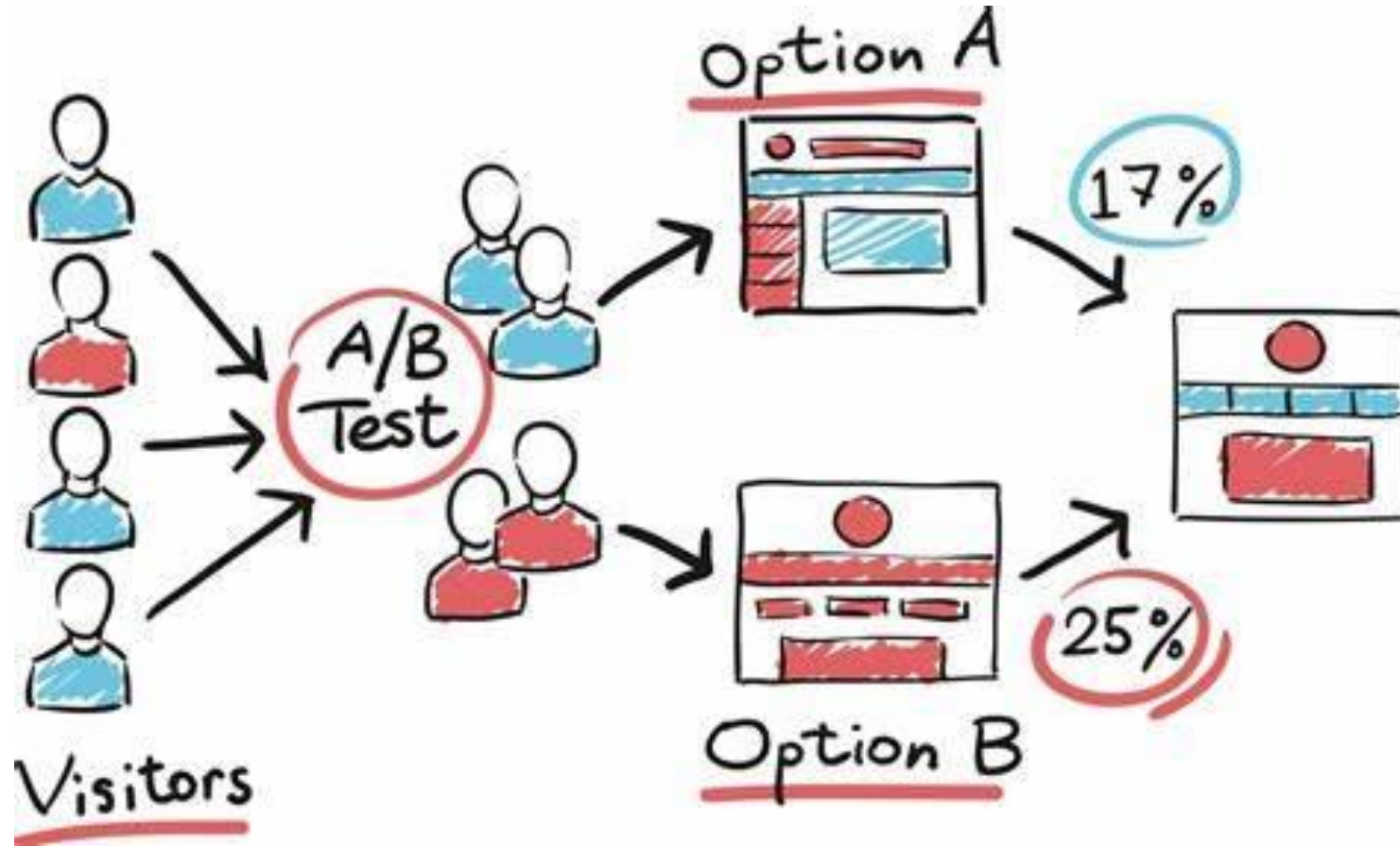


Different versions



- Actual projects have many versions/branches

A/B test (both are in production)



- Different users would see different contents
- Compare the data for version choices

Auto publish by Jenkins



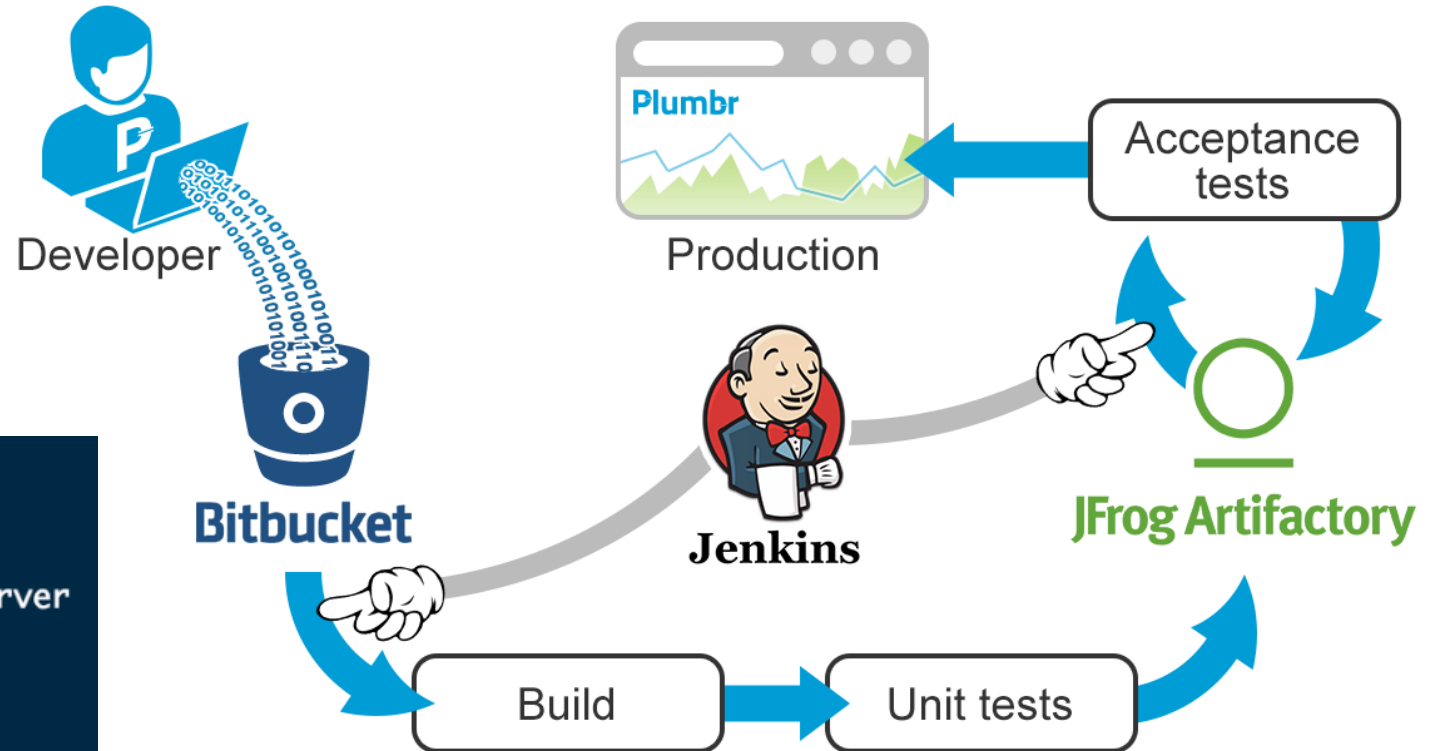
新浪新闻

昨天 15:30 来自 专业版微博平台

+ 关注

【#拼多多回应bug事件#“被薅200亿传言”：实际损失或低于千万元】#拼多多出bug#今天凌晨，拼多多被曝出现重大Bug，用户可领100元无门槛券。网友称“有大批用户开始‘薅羊毛’，一晚上200多亿都是话费充值”。对此，拼多多实际发言人表示，实际最终资损或低于千万元人民币.....
#拼多多回应“被薅200亿”传言：实际损失或低于...

```
root@Dream:~# cd fastjson/class/
root@Dream:~/fastjson/class# ls
ping.class  shell.class
root@Dream:~/fastjson/class# python -m SimpleHTTPServer
Serving HTTP on 0.0.0.0 port 8000 ...
```



- Manually publish project is not a good idea.
- Jenkins is a widely used tool for auto-publish.
- Semi-auto may be a compromise between manual and full-auto.

Operation



客单价 活跃交易用户数 MAU 活动曝光量
用户生命周期价值 用户价值指数
渠道转化率 用户分享率 ROI DAU
用户会话次数 K因子 病毒传播周期 活动参与率
UV 用户获客成本 GMV
用户访问时长 PV 渠道到达量

- Build a website is only the begging.
- Run the business is the core for e-commerce.

Business intelligence



BUSINESS ANALYSIS

- Focuses on processes, techniques and functions
- Used to solve complex business problems and bring out positive change
- Performed by Business Analysts/Systems Analyst/Functional Analyst
- Skills required: Analysis, elicitation, modeling, documentation
- Areas of Impact: Organization, Enterprise, Process, Business and Technology

Pave the way for enabling positive change in an Organization



www.thebusinessanalystsjobdescription.com

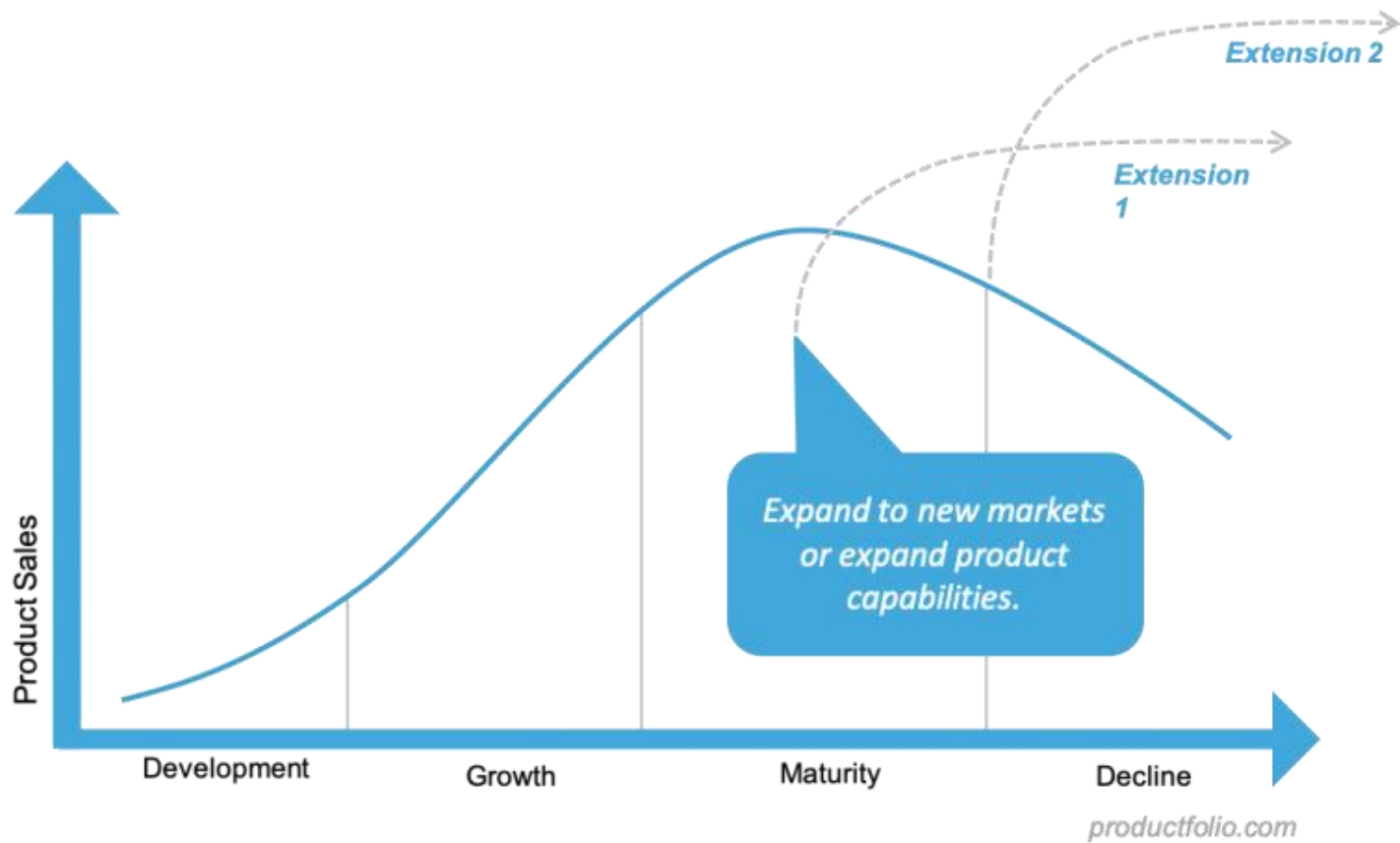
BUSINESS ANALYTICS



- Focuses on data and statistical analysis
- Used to predict future states and drive business decisions
- Performed by Data Analyst, Data Scientist
- Skills required: Statistics, advanced mathematics and software programming
- Areas of Impact: Business decisions, big data, data-driven solutions and forecasting



Product life cycle



Product ≠ Project



In-house or Outsourcing?



In-house team

Advantages

On-site availability	Great expenses
Awareness of the business	Absence issues
In-depth knowledge of the systems & setup	Low retention rates
Deep understanding of the users	Lack of necessary skills



Outsourcing team

Advantages

Access to the talent pool	Management issues
Focus on core business functions	Communication issues
Cost-effectiveness	Problems with quality
Flexibility	
Faster project delivery	



- **Depends on resources**

<https://www.clickittech.com/nearshore/in-house-vs-outsource/>

A miserable story: 7 perpendicular **red** lines



<https://www.bilibili.com/video/BV18s41127AE/>